Proyecto del Plano Inclinado

Determinación teórica y experimental del tiempo de descenso de un bloque con fricción cinética. Reporte generado el 9/10/2026.

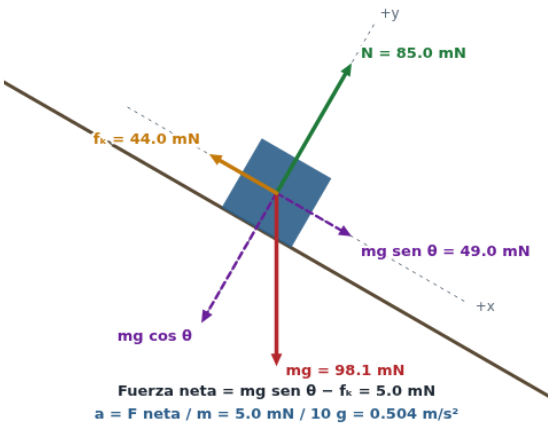
1. Parámetros

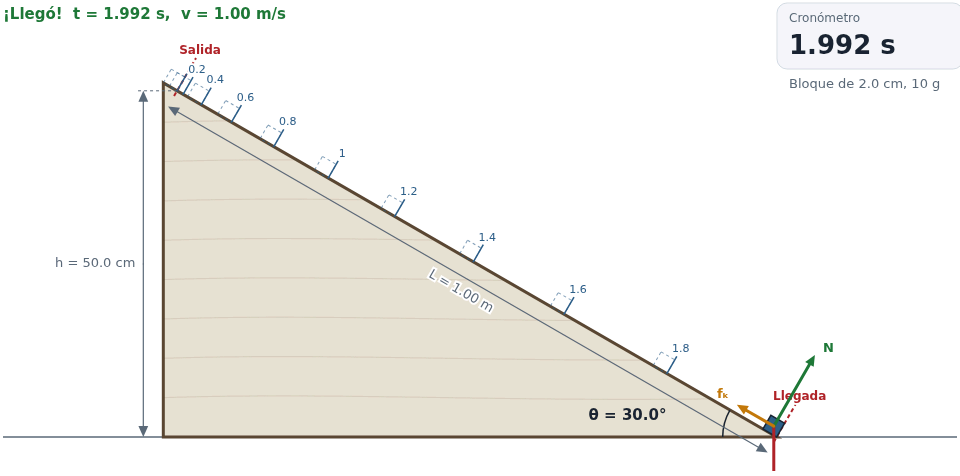
Objeto	Tamaño	m	θ	h	L	μ_k	g	v_0
bloque	2.0 cm	10 g	30°	50.0 cm	1.00 m	0.518	9.81 m/s²	0 m/s

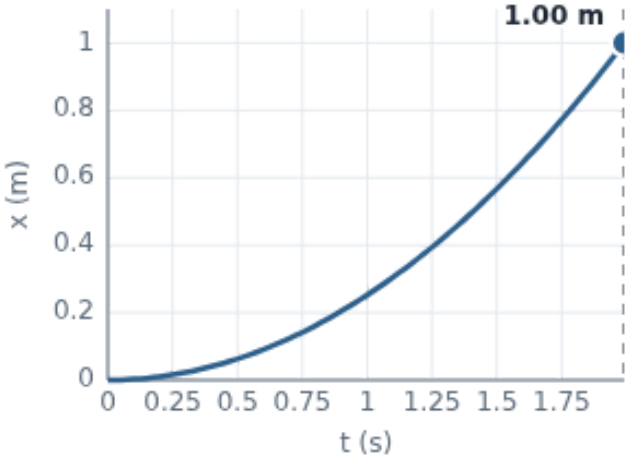
2. Análisis de fuerzas y cálculo teórico

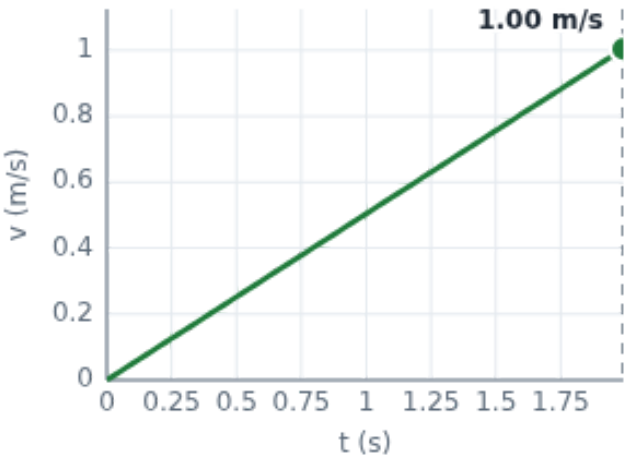
N = mg cos θ; ma = mg sen θ − μ_k mg cos θ ⟹ a = g(sen θ − μ_k cos θ) = 9.81(0.5000 − 0.4486) = **0.504 m/s²**

L = h / sen θ = 50.0 cm / 0.5000 = 1.00 m; t = √(2L/a) = √(2 × 1.000 / 0.504) = **1.992 s**; v = a·t = **1.00 m/s**

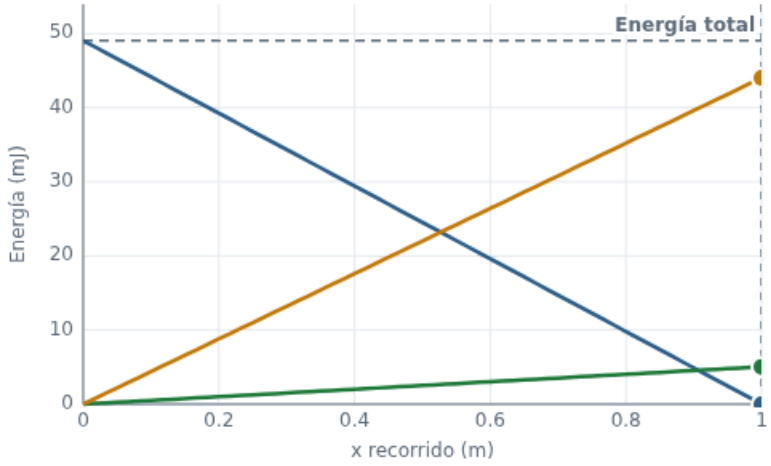





1.00 m
x (m)
t (s)
0
0.2
0.4
0.6
0.8
1
0 0.25 0.5 0.75 1 1.25 1.5 1.75

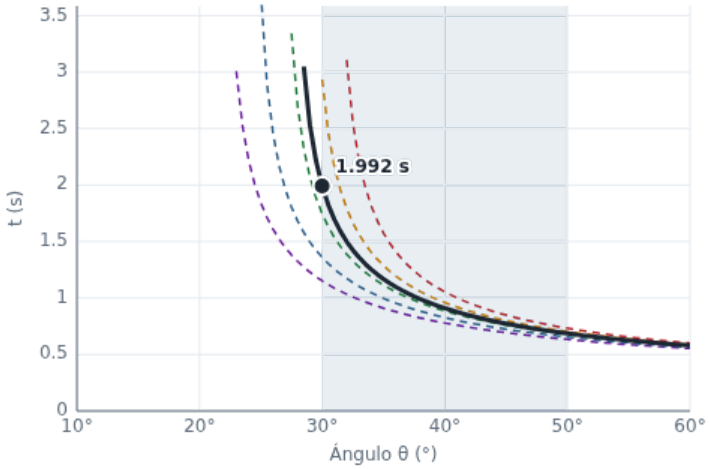

1.00 m/s
v (m/s)
t (s)
0
0.2
0.4
0.6
0.8
1
0 0.25 0.5 0.75 1 1.25 1.5 1.75

3. Energía



4. Sensibilidad (L = 1.00 m)

θ	μ_k = 0.40	μ_k = 0.45	μ_k = 0.50	μ_k = 0.55	μ_k = 0.60
30°	1.152 s	1.360 s	**1.745 s**	2.934 s	no baja
35°	0.911 s	0.997 s	1.115 s	1.287 s	1.576 s
40°	0.779 s	0.827 s	0.886 s	0.959 s	1.055 s
45°	0.693 s	0.724 s	0.759 s	0.800 s	0.849 s
50°	0.633 s	0.654 s	0.677 s	0.703 s	0.732 s



5. Registro experimental

Ensayo	Tiempo medido (s)
1	2.050
2	1.940
3	2.100
4	1.980
5	2.030
Promedio	**2.020**

Error (%) = $|t_{teórico} - t_{experimental}| / t_{experimental} \times 100 = |1.992 - 2.020| / 2.020 \times 100 =$ **1.4 %**

Desviación estándar: 0.062 s. Nota según la escala: **100**. Aceleración medida: 0.490 m/s². μ_k experimental: 0.520.

